

The Anatomy of a Hypothesis

10 Steps to Statistical Truth

1. Formulate Hypotheses

$H_0: \mu = \mu_0$ (Null)

$H_1: \mu \neq \mu_0$ (Alternative)

2. Define Test Statistic

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

3. Set Significance Level

$\alpha = 0.05$ (Threshold)

4. Determine Rejection Region

Rejection Zone
(Area = $\alpha/2$)

5. Collect Data & Calculate

$n = 100$

$\bar{x} = 102.5$

$\sigma = 15$

$Z_{\text{calc}} = 1.67$

6. Plot Calculated Statistic

7. Compare with Rejection Region

8. Make Decision

Since $Z_{\text{calc}} \notin \text{Rejection Region}$,
Do not reject H_0 .

9. Interpret in Context

There is insufficient evidence
to conclude that the mean
differs from μ_0 .

10. Report P-value

p-value = $P(|Z| \geq |1.67|) = 0.095$

p-value > α , therefore the
result is not statistically
significant.

$Z = 1.67$

Non-rejection Zone
(Area = $1-\alpha$)

Hypothesis

A general statement about one or more populations.



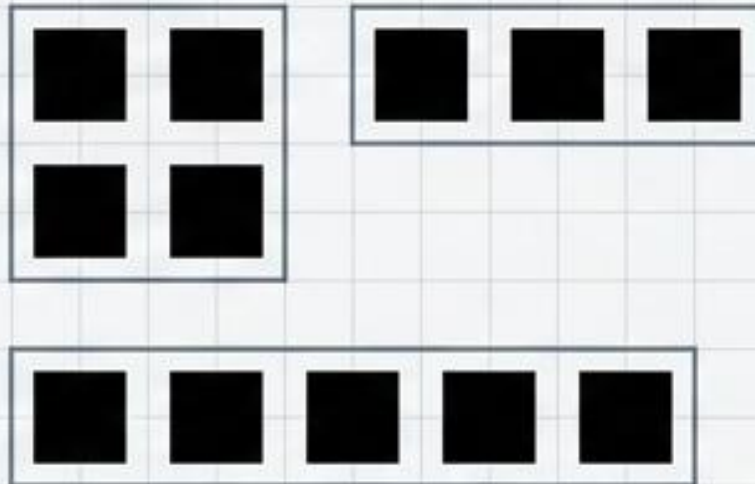
Statistical hypothesis

A hypothesis stated in a way that may be evaluated by appropriate statistical techniques.

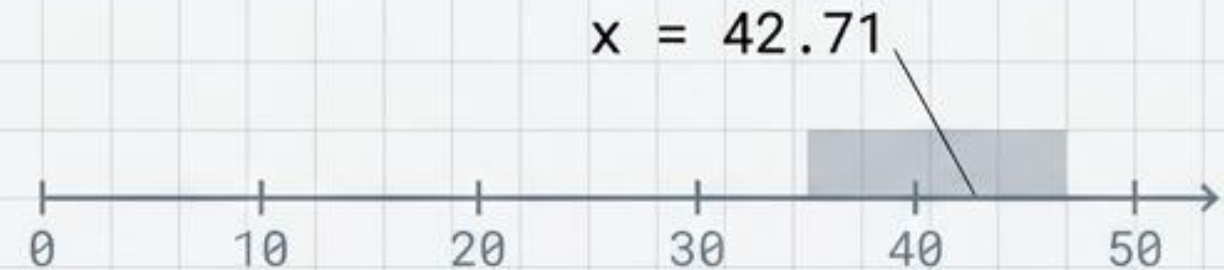
Step 1: Data dictates the test

Every trial begins with raw evidence. The specific type of data collected strictly determines the statistical test that can be used.

Counts
(categorical frequencies)

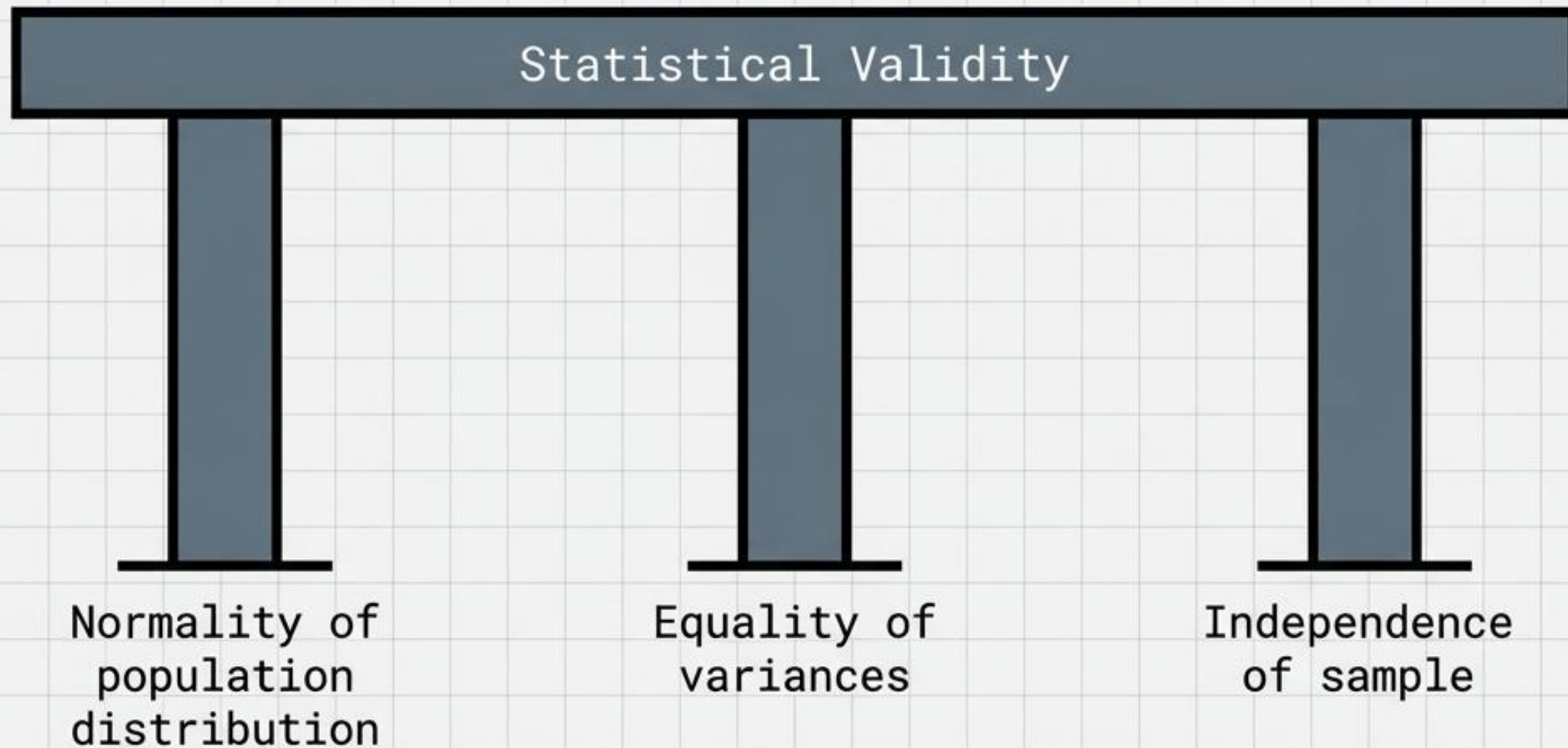


Measurements
(continuous numerical values)



Step 2: Establishing structural assumptions

The test relies on foundational conditions. If these assumptions are not met, the underlying probability distributions collapse.



Step 3: Stating the formal mathematical charges

Null Hypothesis (H_0)

Must contain equality operators ($\leq$, $=$, $\geq$)

$$H_0: \mu = 50$$

$$H_0: \mu \geq 50$$

$$H_0: \mu \leq 50$$



Alternative Hypothesis (H_A)

The strict opposite of the null

$$H_A: \mu \neq 50$$

$$H_A: \mu < 50$$

$$H_A: \mu > 50$$

Step 4: The mathematical standard of proof

Test statistics are computed from the data of the sample and strictly depend on the drawn sample.

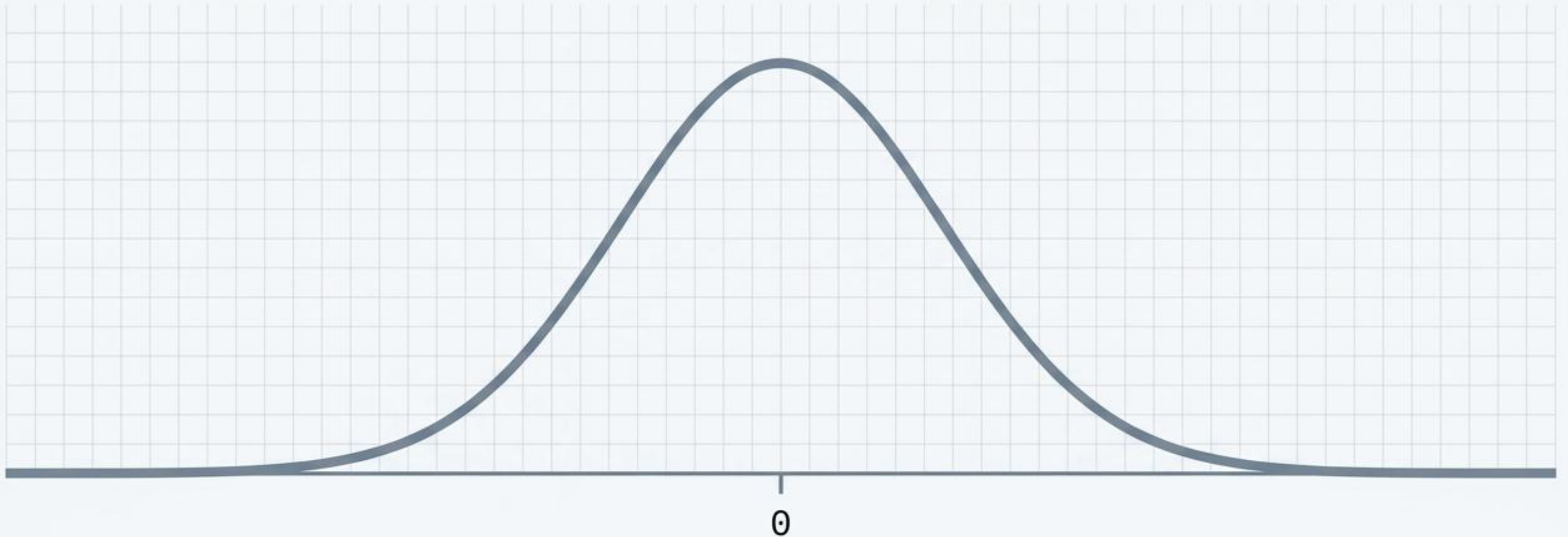
The Signal
The observed effect

$$\text{test statistic} = \frac{(\text{relevant statistic minus hypothesized parameter})}{(\text{standard error of the relevant statistic})}$$

The Noise
The background variance

Step 5: Mapping the probability landscape

This curve represents how the test statistic behaves mathematically if the null hypothesis is true and all assumptions are met.



Standard normal (z)

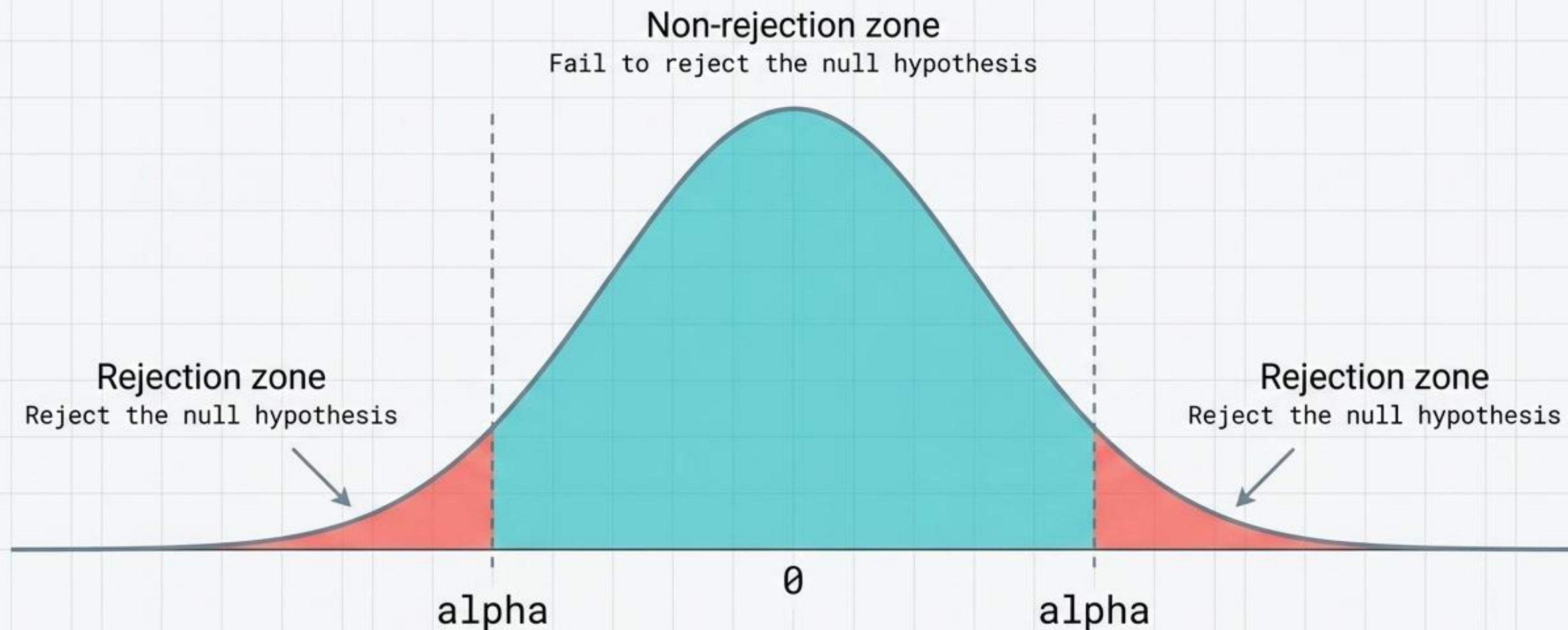
Student's t

F distribution

Chi-square

Step 6: Setting the statistical tripwire

Divide the probability distribution into two distinct groups based on the significance level (alpha).

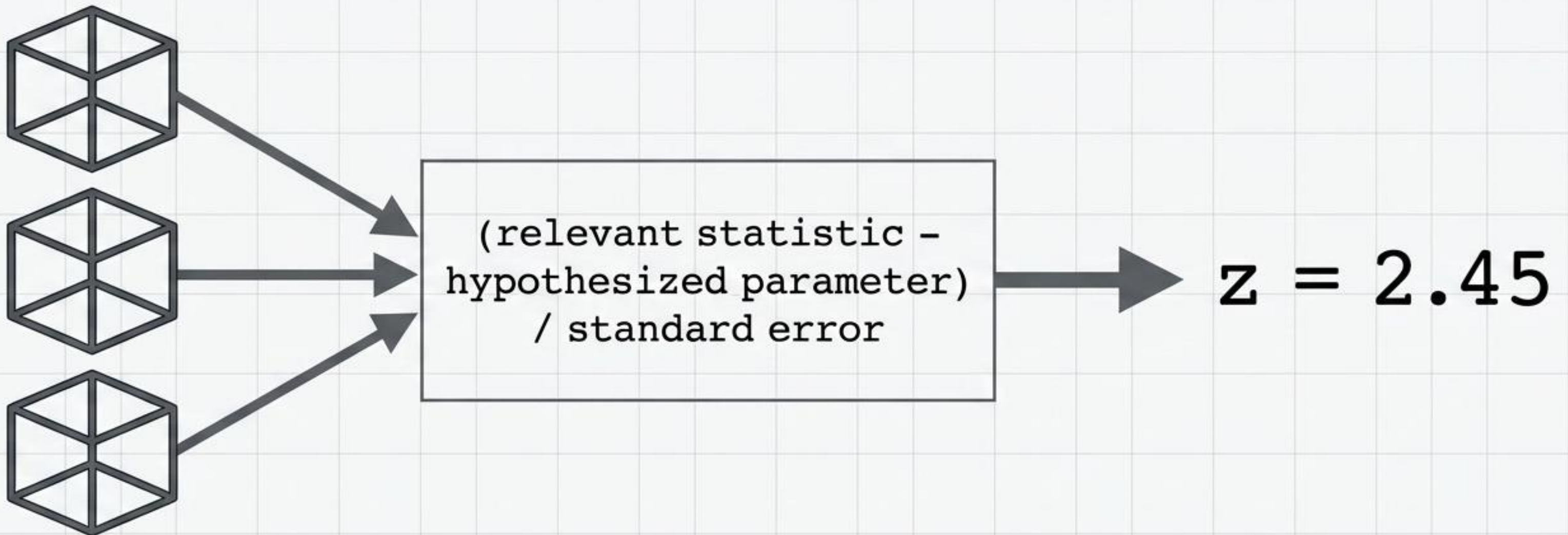


The inherent risks embedded in the decision rule

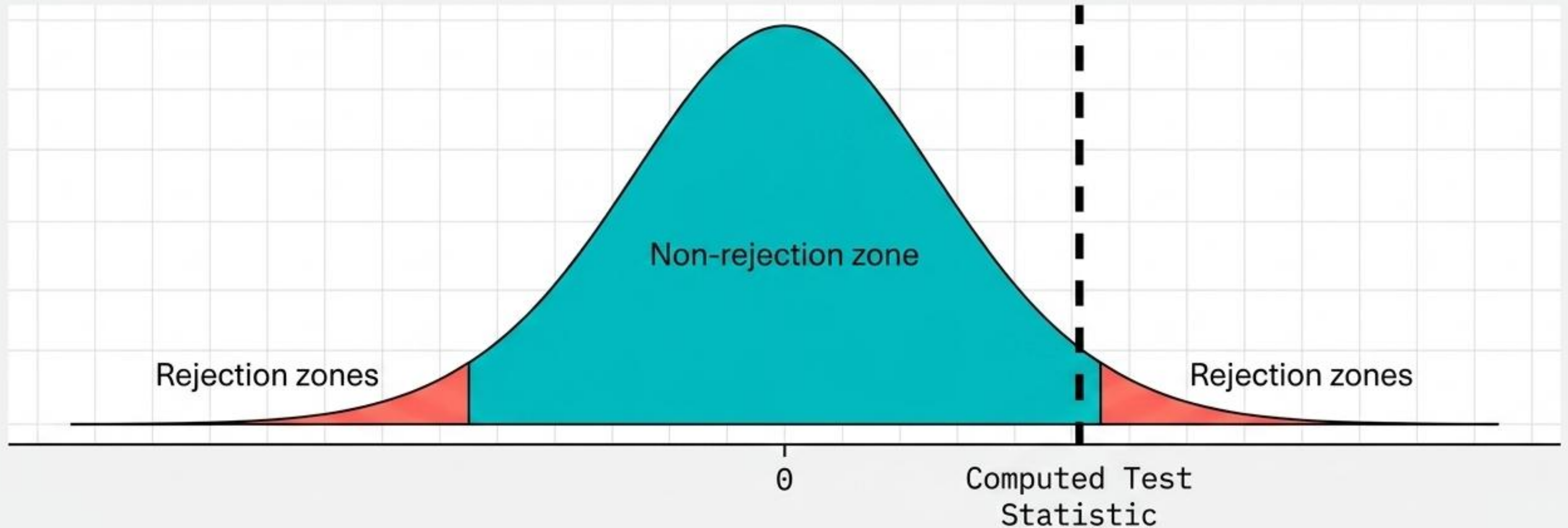
		Reality	
		H0 is True	H0 is False
Decision	Reject H0	Type 1 error Error committed when a true null hypothesis is rejected.	Correct Decision
	Fail to Reject H0	Correct Decision	Type 2 error Error committed when a false null hypothesis is not rejected.

Step 7: Calculating the final test statistic

The relevant statistic, hypothesized parameter, and standard error are combined to produce a single mathematical coordinate.

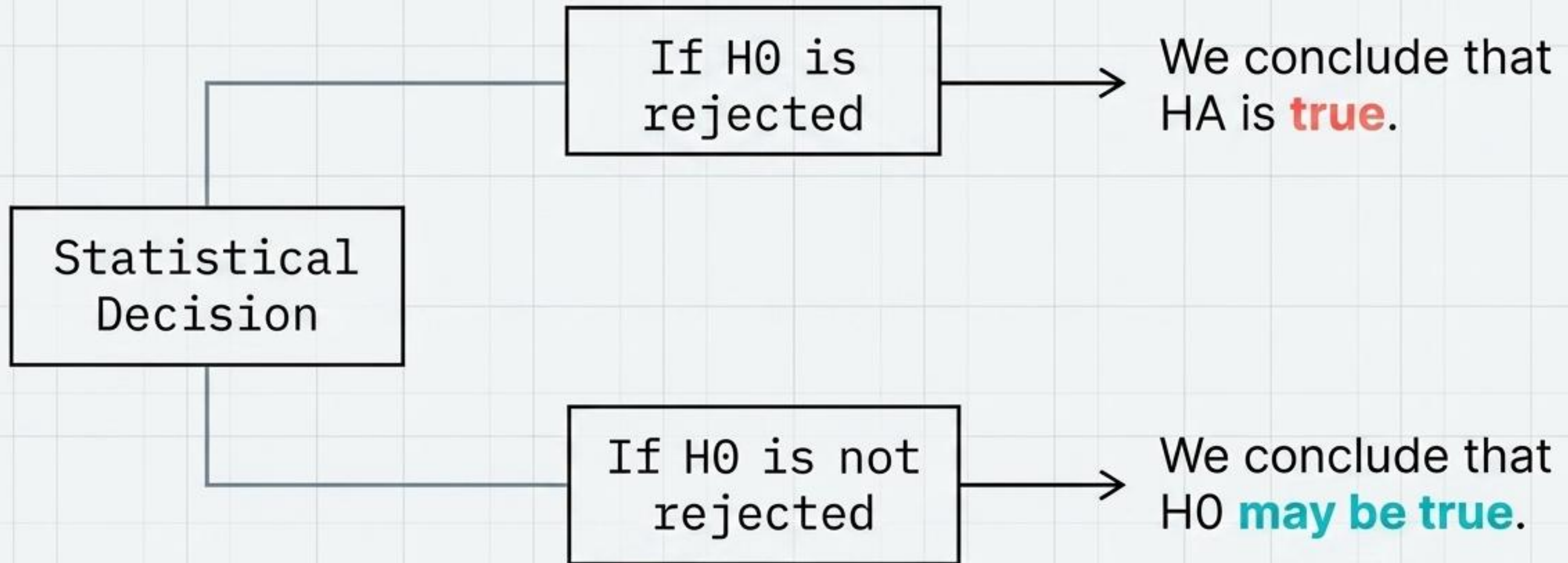


Step 8: Applying the objective ruling



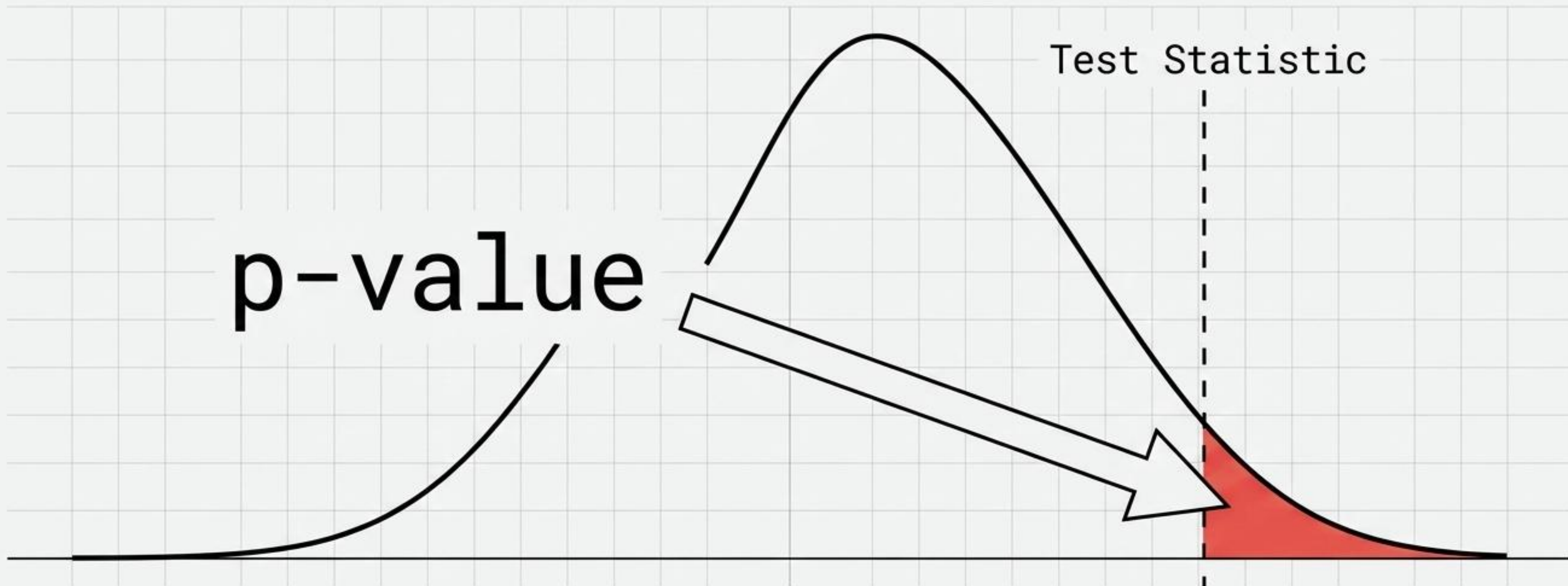
1. Reject the null hypothesis if the computed test statistic is in the rejection zone.
2. Fail to reject the null hypothesis if the computed test statistic is in the non-rejection zone.

Step 9: Translating the math into human reality



Step 10: Isolating the ultimate measure of evidence

The p-value is the smallest value of alpha for which we can reject a null hypothesis when it is true.



The complete statistical framework at a glance

